

IPv6 Implementation -- Guided Notes

Unit 2.3.3 | CompTIA Network+ N10-009 Obj. 2.3 | N10-008 2.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

IPv6 uses _____ -bit addresses, compared to IPv4's 32-bit addresses. Written in hexadecimal, an IPv6 address has _____ groups of four hex digits separated by colons. The total number of possible IPv6 addresses is approximately 3.4×10^{38} (340 undecillion).

Question 2

Two rules for abbreviating IPv6 addresses: (1) Leading _____ in any group may be omitted. (2) One consecutive run of all-zero groups may be replaced with _____. This compression can only be used _____ per address.

Question 3

IPv6 eliminates the need for NAT because every device can receive a globally unique _____ address. This restores the internet's original design principle of _____ connectivity.

Question 4

Match each IPv6 address type to its description:

Global Unicast -> _____

Link-Local -> _____

Multicast -> _____

Loopback -> _____

Question 5

SLAAC stands for _____. It allows IPv6 hosts to configure their own _____ automatically using the network prefix advertised by a router, combined with the host's _____ or a random value -- without needing a DHCP server.

Question 6

IPv6 replaces ARP with _____. It uses _____ messages to discover the link-layer address of a neighbor. This is part of the _____ Discovery Protocol (NDP).

Question 7

During the transition from IPv4 to IPv6, what is a dual-stack configuration? What is tunneling in the context of IPv6 transition, and name one common tunneling mechanism.

Question 8 -- Real World Application

REAL WORLD: A technician pings a server and receives a reply from FE80::1. What does this tell the technician about the current state of connectivity? What would need to be true for the server to be reachable from the public internet over IPv6?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-3-3-ipv6-implementation/2-3-3-ipv6-implementation-answer-key.pdf
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